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Towards Bridging SHAP and Counterfactual Explanations: A Dynamic Aggregation Approach

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Abstract. The widespread applicability of machine learning models across various domains, such as healthcare, agriculture, and finance, has emphasized the need for model explainability due to its black-box nature. Although different neural network-based approaches and other classifiers achieve superior accuracy in classification jobs, they often lack interpretability. Hence, the opacity raises trust issues for these models towards the stakeholders. Shapley Additive Explanations (SHAP) and Counterfactuals are popular explainable methods to address the issue. However, in most research, they are applied independently, considering the distinct perspectives of the models. While SHAP explains why the prediction happened, counterfactual demonstrates what changes in the features make the outcome different. In past research, very limited work has been performed considering this constraint behaviour to build a robust, explainable model. Furthermore, in the case of numerous attributes, stakeholders often need to extract the top k features; hence, understanding these is very important. This study aims to aggregate the two explanations in a dynamic aggregation approach. Experimental results in real-life datasets demonstrate the effectiveness of the proposed method.

Keywords: Explainability, SHAP, Counterfactual, Feature Selection

1 Introduction

In real-world applications, understanding the reason behind the model's prediction is highly important. In various domains such as agriculture [28,20], healthcare [30,3], and finance [4], comprehending the reason is essential to adopt it for the stakeholders. With the emergence of GDPR [14], the development of explainable AI (XAI) is driven by the necessity to comprehend, improve, and build trust issues of models. In the realm of explainability, a spectrum of research efforts has been made to interpret feature importance for a specific prediction. Among them, one of the widely popular methods is SHAP [24], which produces a comprehensive perspective on feature significance based on local explanations. Similarly, another popular method called counterfactual [15,32] is employed to understand how a little modification of a feature can generate the opposite outcome.

During the last decade, there is a growing interest among researchers to illustrate different predictions obtained by various deep learning or machine learning models [29,7]. Although, different explainable models like LIME, GRAD-CAM, etc., provide different and contradictory explanations even when they are applied to the same dataset [25,6]. In recent years, a few important approaches have been available that made an effort to integrate the various individual explainable models to mitigate this problem. For example, in this approach, multiple abductive explanations are aggregated for better explanations [5]. A few recent efforts have been applied to aggregate multiple local explainers [31] hierarchically for generating global explainers. Furthermore, this aggregation cannot be generalized to any type of classification model. In past literature, it is noticed that one primary drawback in SHAP is when a feature is not available, it hinders specifying the model's output [17,16]. SHAP emphasizes the allocation of feature importance scores but fails to provide practical recommendations as it ignores the causal relationships [18,?]. On the contrary, counterfactual explanations yield practical insights while stressing contrastive explanations by introducing a slight change; however, they lack a comprehensive view of feature significance [6]. Moreover, in state-of-the-art research, no strong conclusion is available on how a user may change the behavior of the attribute correctly [22]. Additionally, the work of [16] demonstrates the limitation of SHAP while it is applied alone. Although these two techniques are complementary, nevertheless, a very limited work has been done to integrate these two contradictory approaches for better understanding [2]. In [2], the objective of the method is different as the authors include counterfactuals inside SHAP, and it generates a complex model difficult to understand for the stakeholders. Moreover, testing with different counterfactual approaches is not simple here [2] as the SHAP value is heavily dependent on counterfactual-based background information. This research gap motivates us to formulate explanations

that are interpretable on both global and local levels independently, hence enhancing the applicability of the approach.

This current study generally examines these strategies independently, considering the possible advantages of their amalgamation. Furthermore, in the case of numerous features, it is very important to identify the top k —most features that predict the same. Therefore, in this work, the aim is to propose a method that integrates them efficiently in static and dynamic aggregation approaches. In this paper, experiments are performed on three datasets, e.g., PIMA, water quality, and breast cancer datasets, and the results demonstrate the effectiveness of the proposed idea. We generate the top k —features by the proposed method and then apply traditional classifiers with those feature subsets to validate the model.

The main contributions of the paper are as follows:

- Integration of SHAP and counterfactual sensitivity analysis into a combined framework independent manner, offering both comprehensive and practical local feature importance.
- While combining the effects of the two explainers, static aggregation as well as dynamic aggregation have been performed.
- Proposing a logistic regression-based aggregation method for SHAP and counterfactual sensitivity scores offered a comprehensive understanding of feature importance.
- Validation of the suggested methodology on three datasets illustrates its efficacy in producing dependable and practical outcomes.

The remainder of the article is organized as follows. Section 2 presents the state-of-the-art literature. Section 3 outlines the precursory details about the recent methodology. Section 4 discusses the experimental results, and Section 5 illustrates the experimental results. An attempt to demonstrate the nature of the loss function is kept in Section 6. Finally, Section 7 concludes the article and highlights potential future research directions.

2 Related Work

Over the last decade, significant progress has already been made by employing different deep learning models for the improvement of accuracy in prediction [13]. However, with the growing emphasis on ethical considerations, there is a huge demand for interpretability and explainability, especially across various domains [21]. The goal of Explainable Artificial Intelligence (XAI) is to elucidate the internal decision-making mechanisms of intricate models, thereby improving trust, equity, and transparency [7].

The common objective observed in the past literature is resolving the problem of ambiguous and contradictory explanations obtained from different explainers even for the same instance [25,6]. A general solution is to consider the explanations of multiple explainers for the same instance and then effectively combine them. A study explores the idea of integrating LIME and SHAP by employing logistic regression and random forest algorithms, and they demonstrated the application on healthcare data [1]. In this work, it is assumed that the features are independent of each other, which may not always be true, especially for healthcare data.

Another recent study presents the impact of the Rashomon effect while aggregating the diverse explanations from multiple explainers [25]. In another contemporary work, authors proposed an aggregation model to compare instance-level explanations and it renders humans to understand the bias [19]. In [22], authors only employed the SHAP model considering aggregation as a non-convex optimization problem for interpretability purpose. However, in this work, the integration of SHAP and counterfactuals is not explored.

A recent exploration [23] proposed global aggregations of local explanations (GALE) to address the understanding of how small explanations may provide global insights into a black box model. To solve it, the authors investigated the explanations generated from models trained from a multiclass document classification task and a binary sentiment analysis task with LIME. The research [26] combines multiple local explanations for global explanations. Here, the authors suggested a method to speed up the processing in handling a large-scale data set. Additionally, both lossless and lossy strategies are considered to reduce the time complexity while maintaining the accuracy [9,?].

In some research, authors made an effort to fill the literature gap between feature attribution and counterfactual explanation by mean values. Generally, the research on combinations of SHAP and counterfactuals are explored in two different directions [8]. First, in some other works, researchers demonstrate how feature attribution helps to guide the counterfactual. In contrast, the complementary approach shows that identifying the counterfactuals can strengthen the power of SHAP. A spectrum of work has investigated how to explore counterfactual explanations and feature attribution scores [9].

A recent work [27] presents a method, i.e., AGREE, an aggregation policy that employs weights based on the explanations. This method employs the concept of a neighbourhood-based approach for feature attribution vectors to solve the problems of local and global explainers while demonstrating its robustness in high-dimensional scenarios. Consequently, this can boost the confidence of machine learning models and this is highly needed for trustworthy use in real-world scenarios. However, a very limited approach in integrating explanations with SHAP and counterfactual is present in the current literature. Motivated by this fact, we aim to make the contradictory type of explanations by employing the two and dynamically combining them to generate better explanations.

3 Precursory Details

3.1 SHAP Model

SHAP concept [24] is obtained from *Shapley values*, based on cooperative game theory aiming to explain the output of various machine learning models. It provides an important score for each feature by distributing the feature values.

Suppose that $f(\mathbf{x})$ is the prediction output of the model for a sample $\mathbf{x}$. S be the set of all features, $S = \{x_1, x_2, \dots, x_d\}$ for a model with d features. $S' \subseteq S$ denotes a subset of features removing a feature x_j for which we want to calculate the Shapley value. Finally, $f_{S'}(\mathbf{x})$ is the prediction of the model using only the features in S' .

The mean contribution of x_j over all the subsets of features S' (i.e., subsets of S omitting x_j) generates the Shapley value $\phi_j(f)$ for a feature x_j :

$$\phi_j(f) = \sum_{S' \subseteq S \setminus \{x_j\}} \frac{|S'|!(|S| - |S'| - 1)!}{|S|!} [f(S' \cup \{x_j\}) - f(S')]$$

The term $f(S' \cup \{x_j\}) - f(S')$ signifies the marginal contribution of feature x_j when added to the subset S' . The factorial terms $\frac{|S'|!(|S| - |S'| - 1)!}{|S|!}$ are the *weights* that ensure the average is fairly computed over all possible permutations of features. The sum is taken over all subsets S' of features excluding x_j , and the weight corresponds to the number of permutations where x_j is added after S' .

Counterfactual Explanations

In counterfactual explanations [33], the objective is to identify how a model's prediction changes to a different prediction if some features of $\mathbf{x} = (x_1, x_2, \dots, x_d)$ are modified slightly. Mathematically, the counterfactual explanation is finding a perturbed input $\mathbf{x}'$ such that: $f(\mathbf{x}') \neq f(\mathbf{x})$. Here $f(\mathbf{x})$ is the original model prediction, and $f(\mathbf{x}')$ is the model prediction for the counterfactual input $\mathbf{x}'$.

Given an instance $\mathbf{x}$, the counterfactual explanation aims to find $\mathbf{x}'$ such that: a) the prediction of model changes: $f(\mathbf{x}') \neq f(\mathbf{x})$ and b) the difference between $\mathbf{x}$ and $\mathbf{x}'$ is minimal. This suggests we aim to understand the smallest change of features that causes to alter the prediction.

The perturbation can be performed by considering the distance between the original and counterfactual as a minimization problem. Mathematically, it can be demonstrated as follows:

$$\mathbf{x}' = \operatorname{argmin} \|\mathbf{x} - \mathbf{x}'\|_p \text{ subject to } f(\mathbf{x}') \neq f(\mathbf{x})$$

Here $\|\mathbf{x} - \mathbf{x}'\|_p$ is a distance metric, often the Euclidean distance $\|\mathbf{x} - \mathbf{x}'\|_2$, and the constraint $f(\mathbf{x}') \neq f(\mathbf{x})$ ensures that the prediction changes. For computation purposes, we employ DICE (Diverse Counterfactual Explanations) to generate multiple diverse counterfactual explanations.

4 Proposed Methodology

In this section, we first collected the dataset and three datasets, PIMA, Water quality, and Breast Cancer are considered. Then, we segregated the target from the original dataset, and we divided it into training and test sets. The dataset details are given later.

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4.1 Model Training

Next, we proceed by training a classifier on the dataset. Any classifier, such as a random forest, decision tree, SVM, logistic regression, etc., can be employed. Now, the predictions obtained by the trained model are explained by applying the SHAP and Counterfactual models.

4.2 SHAP and Counterfactual Interpretation

Once the model training is finished, we employ the individual interpretable models like SHAP to find the importance of the feature. After that, we apply the counterfactual to identify the effect of output based on the change of input. For the counterfactual explanation, we compute the sensitivity scores of all the features. The sensitivity score is obtained for multiple test instances selected randomly and then generates a particular n number of counterfactuals for each randomly selected sample. After generating the counterfactual for multiple instances, the sensitivity score can be obtained by **the absolute difference of the original and counterfactual** values of each feature, and the mean value is considered. Now, these SHAP feature importance and the feature importance obtained from sensitivity (i.e., counterfactual) are integrated for a better explanation.

4.3 Static Aggregation of Individual Models

To combine the explanations of two individual models, we average the features obtained from both the individual explainers, i.e., the importance obtained from SHAP and the sensitivity score received from the counterfactual. Now the validation is performed for the static aggregation.

4.4 Dynamic Aggregation of Individual Model

To aggregate the predictions of two individual models dynamically and improve the overall performance, a logistic regression model is employed. Here, for the training purpose, the feature importance obtained from SHAP and the sensitivity score obtained from counterfactual are combined to plug into as the input of the classifier. Here, the same target is used for the training purpose. Once the training is finished, the corresponding weights for each feature are incorporated to find the final importance considering both the conflicting criteria.

Note that, for SHAP, the mean absolute importance for each feature is considered, and the SHAP values are obtained for all the training samples. However, for the sensitivity score obtained from counterfactuals, the mean feature importance (across some randomly taken instances) is replicated for all the training samples to make an alignment with SHAP. Considering multiple instances for generating the diverse set of counterfactuals ensures robustness.

4.5 Weighted Sum Formula

The combined feature importance for various features $\mathbf{C}$ is expressed as:

$$\mathbf{C} = \mathbf{w}_{\text{SHAP}} \cdot \mathbf{x}_{\text{SHAP}} + \mathbf{w}_{\text{CF}} \cdot \mathbf{x}_{\text{CF}}$$

Where the individual components C_i are:

$$C_i = w_{\text{SHAP},i} \cdot x_{\text{SHAP},i} + w_{\text{CF},i} \cdot x_{\text{CF},i}$$

Here, w_{SHAP} and w_{CF} denote the weights for the SHAP and counterfactual obtained from the logistic regression. On the other hand, x_{SHAP} and x_{CF} are the feature importance obtained from SHAP and counterfactual, respectively.

4.6 Validation of the Model

For validation purposes, the top k features are selected from the dataset **based on the importance obtained from the three individual models**. Then, **the** same classifier is employed, and the target is predicted. Here, the accuracy and F1 score are computed for comparison purposes among all the individual explainers along with the proposed method.

5 Experimental Design and Analysis

To perform the experimental study of the work, we consider the three datasets, e.g., PIMA [11], Water Quality [12], and Breast cancer dataset. The breast cancer dataset is available publicly in the Scikit-learn [10].

5.1 Brief Description about the Dataset

This PIMA dataset compiles the medical information from the female patients, and the objective is to forecast the possibilities of diabetics based on the medical information. This dataset has 768 rows and 9 features. The features are “pregnancies”, “glucose”, “bp”, “skin thickness”, “insulin”, “bmi”, “pedigree”, “age”, “outcome”. Similarly, the water quality dataset [12] has 10 features like PH, Chloramines, Hardness, etc., and the target is Potability (either 0 or 1). The sample size is 3276. The breast cancer dataset has 569 samples with 30 attributes, and it has the binary target.

5.2 Results Obtained from Individual Explainer

Initially, the dataset is split into an 80:20 ratio for training and testing purposes. Then random forest is applied and SHAP values for the whole training set are obtained. Finally, the absolute feature importances for a particular class are considered for SHAP. For diverse counterfactual generation across the datasets, we selected 10 random test samples, and for each sample, 5 counterfactual examples were computed, and the average values are considered. Then, these two sets of feature importance are combined horizontally. Thus, it serves as the input of the logistic regression model to derive the weights of each feature across the two explanations.

The results obtained for a random sample collected for the 5 counterfactual explanations on PIMA is depicted in Fig. 1. It can be noticed that, although for this sample the target outcome is 0, but if the changes are made jointly with bmi and glucose, the outcome can be altered.

	pregnancies	glucose	bp	skin_thickness	insulin	bmi	pedigree	age	outcome
0	1	80	55	0	0	19.1	0.258	21	0
Diverse Counterfactual set (new outcome: 1)									
	pregnancies	glucose	bp	skin_thickness	insulin	bmi	pedigree	age	outcome
0	1	154	55	0	0	43.9	0.258	21	1
1	1	166	55	0	0	51.6	0.258	21	1
2	1	188	55	0	0	36.3	0.258	21	1
3	1	158	55	0	0	64.4	0.258	21	1
4	1	167	55	0	0	35.5	0.258	21	1

Fig. 1: A snapshot of counterfactual explanation on PIMA.

After applying the individual model, the experimental results obtained from SHAP feature importance and the sensitivity score collected from DICE are plotted in the same diagram. The Fig. 2 demonstrates that for SHAP, glucose, BMI, age, and pregnancies are the top-most features, whereas, for counterfactual interpretation, insulin, bpm glucose, and skin thickness become the highly important features. Here, it can be easily identified that in some features, the diverse explanations are obtained from these two explainers. To exemplify, for insulin, it can be noticed that SHAP demonstrates less importance, while counterfactual denotes high importance. Similarly, for pedigree, the feature has high importance demonstrated by SHAP, but it has less importance by counterfactual.

Next, we demonstrate the experimental results by the average of SHAP and the sensitivity scores obtained from the counterfactual. In this experimental study, as random forest is applied, therefore, the feature importance obtained from the classifier itself is also computed. In Fig. 3, the diverse plots of various explainers, along with the combined feature importance obtained from SHAP and counterfactual explanations, are demonstrated. Before making the combination, the values are normalized to make at comparable. From this figure, it can be noticed that, for example, for age, the importance of SHAP is very high while the sensitivity has a low value, but the combined effect of this tends the overall effect to reduce and it becomes close to the actual importance. Similarly, for insulin, it can be noticed that sensitivity has a very high value, while SHAP has lower importance. Here, although random forest has its own feature importance property, global interpretability and providing actionable insights by perturbing

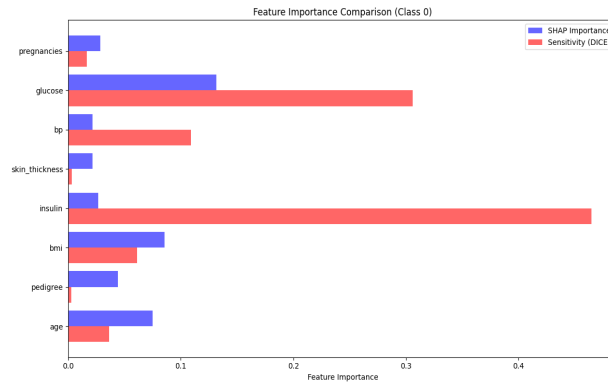


Fig. 2: Feature importance for SHAP and Sensitivity scores from counterfactual.

the features are not possible without counterfactual, and hence, combinations of these two explainers provide meaningful insights. However, instead of static averaging, the feature importance score for PIMA after dynamic aggregation is demonstrated in Fig. 4.

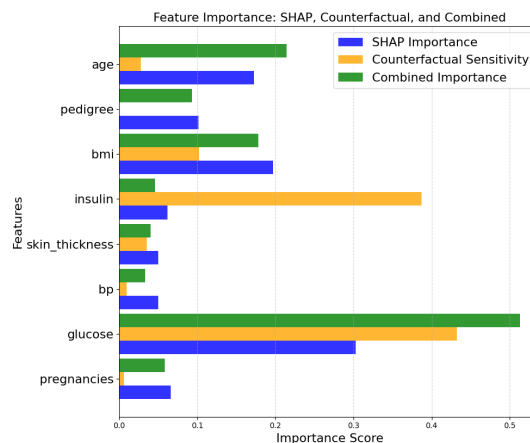


Fig. 3: Demonstration of feature importance when average combination is considered.

To perform the validation of the model top- K features are selected for various explainers. Here, for the experiment purpose, we consider the top 3,4,5,6,7 features, respectively, and then apply random forest on the subset of features. To evaluate the performance, accuracy and an F1 score are chosen and the comparative results for PIMA dataset are reported in Table 1. It can be noticed that the majority of times, the combined performance is better than the counterfactual, while for $K=3$ and 5, the combination illustrates the effectiveness when compared to SHAP. Similarly, SHAP outperforms the combination approach for $K=4$ and 6, however, relying only on SHAP it cannot produce robust explanations all the time. The experimental findings obtained for the other two datasets, i.e., water quality and breast cancer detection, are presented in Tables 2 and 3, respectively. For the water quality dataset, random forest is considered, whereas the SVM classifier is employed for the breast cancer dataset. In the breast cancer dataset, for the training, SVM with radial basis function as kernel is applied, and 9 counterfactuals are generated for 10 random samples. However, in the last phase, for validation purposes, we employ various versions of SVM, and the results for linear SVM are reported. Filter and wrapper methods are also popular feature selection models, however, they lack interpretability criteria. Therefore, from Tables 2, it is noticed that in the majority of cases proposed method performs better than the counterfactual, and it is competitively good with SHAP. Table 3 reflects that combined explainer perform better or avoids worst in the majority times than individual explainers. Hence, altogether, this demonstrates the effectiveness of the proposed method for proper feature selection considering interpretability and accuracy issues.

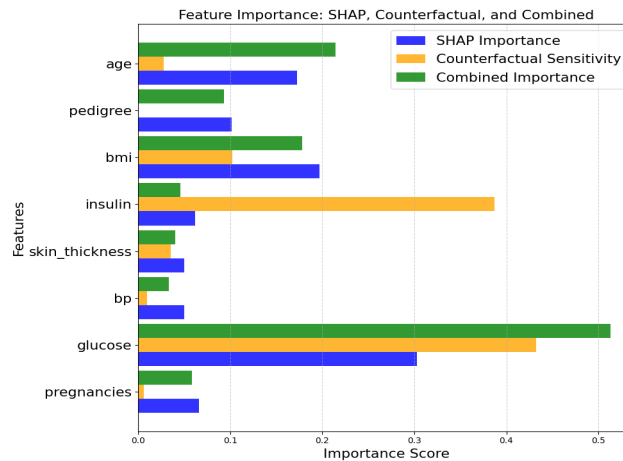


Fig. 4: Feature importance obtained after dynamic aggregation on PIMA.

Table 1: Comparative analysis on PIMA dataset.

Top-K features	Accuracy (SHAP)	F1 Score (SHAP)	Accuracy (Sensitivity)	F1 Score (Sensitivity)	Accuracy (Combined)	F1 Score (Combined)
3	0.720779	0.612613	0.681818	0.566372	0.733766	0.630631
4	0.766234	0.684211	0.701299	0.596491	0.753247	0.666667
5	0.772727	0.690265	0.727273	0.644068	0.779221	0.701754
6	0.772727	0.690265	0.720779	0.626087	0.766234	0.678571
7	0.759740	0.678261	0.720779	0.612613	0.759740	0.683761

6 Theoretical Insights

Lemma 1. *Presence of optimal weights*

Proof (Statement). Considering binary classification outcomes $y \in \{0, 1\}$ and features S and C derived from two independent explainers, there exist optimal weights w_S, w_C that minimize the logistic loss function:

$$L(w) = - \sum_{i=1}^n \left[y_i \log \sigma(w_S S_i + w_C C_i) + (1 - y_i) \log(1 - \sigma(w_S S_i + w_C C_i)) \right],$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. Here we first need to define the logistic loss as follows:

$$L(w) = - \sum_{i=1}^n \left[y_i \log \sigma(w^T X_i) + (1 - y_i) \log(1 - \sigma(w^T X_i)) \right],$$

where $w = [w_S, w_C]^T$ and the combination vector of explanation is $X = [S, C]$. Now we need to prove that logistic loss is a convex function of w considering the second derivative matrix, i.e., the Hessian of $L(w)$ is positive semi-definite.

Gradient of Logistic Loss

We aim to derive the gradient of the logistic loss function concerning the weights w .

Define the components in simple terms

Let

$$z = w^T X, \quad \hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

and the logistic loss for one data point is:

$$\ell(w) = - [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

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Table 2: Comparative analysis on Water quality dataset.

Top K features	Accuracy (SHAP)	F1 Score (SHAP)	Accuracy (Sensitivity)	F1 Score (Sensitivity)	Accuracy (Combined)	F1 Score (Combined)
2	0.617378	0.445916	0.606707	0.431718	0.617378	0.445916
3	0.652439	0.474654	0.612805	0.386473	0.652439	0.474654
4	0.637195	0.430622	0.620427	0.372796	0.678354	0.505855
5	0.678354	0.484108	0.609756	0.329843	0.684451	0.488889
6	0.685976	0.471795	0.653963	0.374656	0.673780	0.478049
7	0.682927	0.469388	0.632622	0.360743	0.679878	0.464286
8	0.673780	0.436842	0.672256	0.441558	0.681402	0.462725

Table 3: Comparative analysis Breast Cancer dataset.

Top K features	Accuracy (SHAP)	F1 Score (SHAP)	Accuracy (Sensitivity)	F1 Score (Sensitivity)	Accuracy (Combined)	F1 Score (Combined)
5	0.964912	0.972222	0.973684	0.979310	0.964912	0.972603
6	0.964912	0.972222	0.964912	0.972603	0.964912	0.972603
7	0.956140	0.965517	0.964912	0.972603	0.964912	0.972603
8	0.964912	0.972222	0.973684	0.979310	0.964912	0.972603
9	0.964912	0.972222	0.973684	0.979021	0.973684	0.979310
10	0.964912	0.972222	0.956140	0.965517	0.973684	0.979310
11	0.964912	0.972222	0.956140	0.965517	0.973684	0.979310
12	0.964912	0.972222	0.956140	0.965517	0.964912	0.972603
13	0.964912	0.972222	0.956140	0.965517	0.973684	0.979310
14	0.964912	0.972222	0.956140	0.965517	0.973684	0.979310

Derivation of Gradient and Hessian for Logistic Loss

We define the logistic loss function over n samples as:

$$L(w_S, w_C) = - \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)],$$

where:

$$z_i = w_S S_i + w_C C_i, \quad \hat{y}_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}.$$

First Derivative (Gradient)

Using the chain rule, we compute the first partial derivatives:

$$\begin{aligned} \frac{\partial L}{\partial w_S} &= \sum_{i=1}^n \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial z_i} \cdot \frac{\partial z_i}{\partial w_S} = \sum_{i=1}^n (\hat{y}_i - y_i) S_i \\ \frac{\partial L}{\partial w_C} &= \sum_{i=1}^n (\hat{y}_i - y_i) C_i \end{aligned}$$

Second Derivative (Hessian)

We now compute the second-order partial derivatives.

Second Derivative w.r.t. w_S

$$\frac{\partial^2 L}{\partial w_S^2} = \sum_{i=1}^n \frac{d}{dw_S} [(\hat{y}_i - y_i) S_i] = \sum_{i=1}^n \hat{y}_i (1 - \hat{y}_i) S_i^2$$

Second Derivative w.r.t. w_C

$$\frac{\partial^2 L}{\partial w_C^2} = \sum_{i=1}^n \hat{y}_i(1 - \hat{y}_i)C_i^2$$

Mixed Partial Derivative

$$\frac{\partial^2 L}{\partial w_S \partial w_C} = \sum_{i=1}^n \hat{y}_i(1 - \hat{y}_i)S_iC_i$$

Hessian Matrix

The Hessian matrix H is given by:

$$H = \nabla^2 L = \sum_{i=1}^n \hat{y}_i(1 - \hat{y}_i) \begin{bmatrix} S_i^2 & S_iC_i \\ S_iC_i & C_i^2 \end{bmatrix}$$

In this Hessian matrix, each term in the matrix is positive, hence this matrix is semi-definite. Thus it demonstrates that the logistic loss is convex.

7 Conclusion

In this paper, the integration of SHAP and counterfactuals is performed to find a proper subset of features. We apply a logistic regression-based dynamic aggregation to find the weights. Here, the objective of the study is to derive the top k features among multiple attributes and train the model with the subset of features. Finally, the validation is measured based on accuracy and F1 scores. The experimental outcomes exhibit that majority of cases, the proposed method performs well, while in some cases it is competitively good with SHAP. Hence, it can be effective for the stakeholders while considering relevant interpretable features without losing accuracy. In future, more rigorous analysis should be performed on other datasets along with integrating other explainable models that consider local and global aspects with fairness.

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